



Cooling behaviors of liquid hydrogen by helium gas injection

Fushou Xie¹ · Yanzhong Li^{1,2,3} · Kang Zhu¹ · Yuan Ma¹ · Jiaojiao Wang¹

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Abstract

The cooling method of helium bubbling is of great importance in the acquisition system of densified liquid hydrogen because of its safety, reliability and easy operation. To obtain better cooling effect in the cooling system, in this paper, a mathematical model of instantaneous heat and mass transfer is developed by using two-film theory to describe the cooling behavior of a single helium bubble. The solubility and diffusion processes for the single helium bubble injected into liquid hydrogen are quantitatively analyzed. The dominant factors affecting the cooling capacity are obtained on the basis of discussion on several influencing factors. Taking these as important basis of the assumption, a thermodynamic model considering effects of static pressure and thermo-physical properties is established to accurately predict the temperature dropping process of liquid hydrogen by helium bubbling. Compared with the associated experimental data, it is found that the calculating results using the thermodynamic model proposed in the present paper are well consistent with the experimental data. The cooling effect becomes better with the decreases of ullage pressure, injected helium temperature and environmental heat flux. The helium gas consumption gradually reduces with the increase of helium injection rate and the decrease of environment heat flux. Meanwhile, the same method is also used for the cooling of liquid oxygen, liquid nitrogen and liquid methane. It shows that the cooling effect using the method of helium gas bubbling varies for different cryogenic liquids. The triple-point temperatures of LN₂ and LCH₄ can easily be reached by the helium injection, but it is hardly cooled down to their freezing points for the LH₂ and LO₂. Among the four cryogenic liquids, the LH₂ cooling needs the largest consumption of GHe to meet the requirement of a certain subcooling degree.

Nomenclature

A	Surface area, m ²
c_p	Specific heat, kJ/(kg K)
C	Concentration, mol
d	Diameter, m
D	Mass diffusion coefficient, m ² /s
E	Cooling capacity, J
g	Gravitational acceleration, m/s ²
h	Convective heat transfer coefficient, W/(m ² K)
H	Enthalpy, J/kg
i	Grid node, $i = 0, \dots, n$
k	Mass transfer coefficient, m/s

k_B	Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$, J/K
L	Liquid level, m
m	Mass, kg
$\dot{m}$	Mass flow rate, kg/s
M	Mole mass, kg/kmol
n	Mole number, mol
p	Pressure, Pa
p^*	Dimensionless pressure
q	Total sensible heat flux density, W/m ²
Q	Heat flux, W
R_g	Universal gas constant
R_o	Helium molecule radius
T	Temperature, K
u	Bubble terminal velocity, m/s
V	Volume, m ³
$\dot{V}$	Volumetric flow rate, m ³ /s
x	Mole fraction
X	Mole ratio
Z	Compressibility factor

✉ Yanzhong Li
 yzli-epe@mail.xjtu.edu.cn

¹ Institute of Refrigerating and Cryogenic Engineering, Xi'an Jiaotong University, Xi'an 710049, China

² State Key Laboratory of Technologies in Space Cryogenic Propellants, Beijing 100028, China

³ School of Energy and Power Engineering, Xi'an Jiaotong University, Xi'an 710049, China

Greek symbols

σ	Surface tension, N/m
δ	Thickness, m
λ	Thermal conductivity, W/(m K)

$erf(\xi)$	Gaussian error function
τ	Time, s
$d\tau$	Time step, s
μ	Dynamic viscosity coefficient, Pa s
ρ	Density, kg/m ³
ν	Kinematic viscosity, m ² /s
α	Thermal diffusion coefficient, m ² /s
ω	Diffusion volume of molecule, m ³
γ	Latent heat, J/kg
φ	Gas phase fraction
θ	Excess temperature
ψ	Helium mass ratio
Nu	Nusselt number, $Nu = hd/\lambda$
Re	Reynolds number, $Re = ud/\mu$
Pr	Prandtl number, $Pr = \mu c_p/\lambda$
Le	Lewis number, $Le = \alpha/D$

Abbreviations and subscripts

a	Required helium consumption at cryogenic liquids cooling to any temperature
b	Bubble
boil	Normal boiling
c	Latent heat flux
d	Dissolution
diff	Diffusion
e	Environment
f	Film temperature, $T_f = (T_{LH_2} + T_b) / 2$
G	Vapor film
i	Interface 'i'
in	Flowing into the fictitious gas phase
inside	On the inside of a single bubble
u	Liquid film
m	Mixture of hydrogen-helium
n	Required helium consumption at cryogenic liquids cooling to the minimum attainable temperature
o	Ullage
out	Outflowing from the fictitious gas phase
outside	On the outside of a single bubble
REF	Reference pressure
s	Interface 's'
sat	Saturation
t	Total
triple	Triple point
u	Liquid film
H ₂	Hydrogen
He	Helium
LH ₂	Liquid hydrogen
GH ₂	Hydrogen gas
LHe	Liquid helium
GHe	Helium gas
LO ₂	Liquid oxygen
LN ₂	Liquid nitrogen
LCH ₄	Liquid methane

1 Introduction

Helium gas (GHe) bubbling is a cooling method of obtaining subcooled liquid hydrogen (LH₂), as shown in Fig. 1, which is beneficial to cryogenic rocket. The cooling mechanism of helium bubbling is that hydrogen molecules are driven to migrate into helium bubbles by concentration difference. Due to the absorption of heat of vaporization from LH₂, the cooling capacity is generated to keep LH₂ temperature below its saturation temperature.

This cooling method has been used to provide subcooled liquid oxygen (LO₂) in the suction lines of the Saturn booster and in several missile programs during ground prelaunch [1, 2]. Because of ambient heating, the LO₂ temperature at the turbo-pump inlet approaches its saturation state, possibly resulting in the failure of the engine during start, so subcooled cryogenic propellants are chosen [3]. Recently, subcooled cryogenic propellants have been obtained not only to meet entrance temperature requirements of turbo-pump but also to improve performance of launch vehicles [4]. For instance, when the LH₂ temperature decreases from the normal boiling point temperature (20.39 K) to its triple-point temperature (13.8 K), its density and sensible cooling capacity per unit volume will increase by 8.8% and 20%, respectively [5]. If the densities of LH₂ and LO₂ increase by 8% and 10%, respectively, the total take-off weight of launch vehicles can be reduced by 20% [6]. Meanwhile, the GHe injection is also applied in the cryogenic propellant pressurization for rocket upper stages because of its simplicity [7, 8]. Consequently, the GHe injection system is simpler and more reliable than a hot-gas system where the GHe is injected through the overhead injector at high temperature, as shown in Fig. 2.

For these applications, several investigators [1–3, 7–14] have conducted studies on heat and mass transfer processes of GHe injection in cryogenic propellants. Schmidt [1] experimentally investigated the cooling effect of LH₂ by GHe injection method. It is indicated that the cooling of cryogenic liquids by injecting a non-condensable gas into the liquids is feasible. Larsen [3] proposed a theoretical model to account for the cooling behaviors, and the maximum attainable

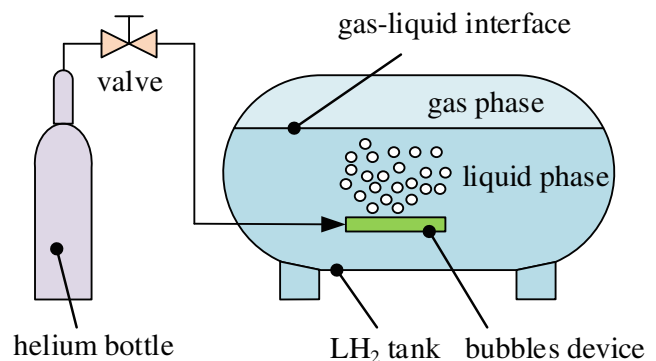


Fig. 1 Sketch of helium gas injection method

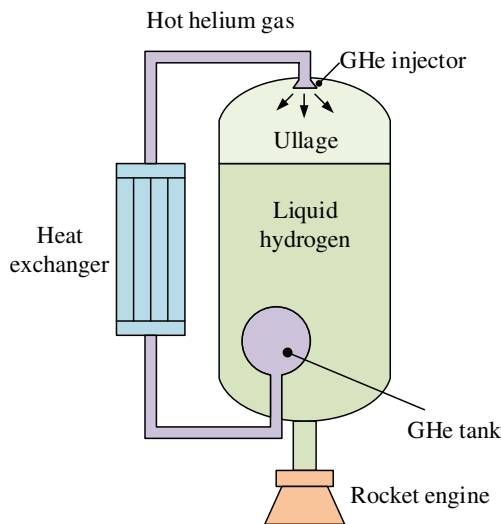


Fig. 2 Sketch of hot gas pressurization system

subcooling can be obtained by this analysis. Lytle [9] presented some experimental data on the cooling effect of various liquids through injecting GHe. The results indicated that GHe injection method to keep temperature below the boiling point of the coolant is simpler than the usual vapor-pumping technique. Ramesh [10] also indicated that the subcooling technique using helium gas injection is a very simple method. It can be very well adopted in propellant tanks for ground applications of launch vehicles. Cho [11, 12] developed the finite rate heat transfer and instantaneous mass transfer model, which is in agreement with experimental data for high flow rate of helium injection ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.39 \text{ min}^{-1}, 2.07 \text{ min}^{-1}$), but inconsistent for low flow rate of helium injection ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 0.85 \text{ min}^{-1}$). Jung [13] reported that the dynamics of mass transfer is dominant and the transient heat transfer is negligible during the helium injection process. Sandilya [14] proposed a lumped parameter model considering heat and mass interaction between bubbles and liquid to gain a preliminary insight into the cooling phenomenon during helium gas injection.

Moreover, much work has been carried out on the heat and mass transfer for cooling behaviors [15–27]. Kilic [15] investigated the heat transfer from a porous plate by cooling of air with transpiration. The results show that increasing Reynolds number causes an increase on surface temperature and a decrease on cooling effectiveness of porous plate and efficiency of the system. Liu [19] developed a general mathematical model to simulate simultaneous transfer of heat and mass in the porous media unsaturated with liquid. Chen [20] studied the supercooling phenomenon and the freezing probability of water inside horizontal cylinder during a cooling process of thermal storage. The results show that the lower the coolant

temperature, the greater the probability of nucleation. Zhao [21] focused on the numerical simulation and experimental study of heat and mass transfer in fuel droplet evaporation. Results show that when the ambient temperature is relatively low, the diesel fuel droplets first pass through the initial heating stage and then quickly evaporate. Liu [23] conducted numerical simulation and experimental investigation with distilled water as the test liquid to study the spray cooling heat transfer characteristics in the non-boiling region. It is found that droplet diameter is an important influence factor on spray cooling characteristics in the non-boiling region, especially at low heat flux. Kumano [26] investigated experimentally the flow characteristics of ice slurry during cooling. The flow and solidification characteristics of the ice slurry were found to be dependent on the flow rate and ice packing factor.

From the existing investigations, it can be seen that the cooling method by injecting GHe into liquids is complex heat and mass transfer processes, and the fluid flow, heat transfer, and mass transfer are coupled strongly. Although much work has been done on the cooling method by using GHe bubbling, few researchers have proposed a reasonable theoretical model to accurately describe the cooling behaviors, and no work has been published on the discussion of key factors affecting cooling effect, such as thermophysical properties, system pressure and so on. In the open literatures, the core issues concerned in practical applications are also not reported, which include minimum attainable temperature, GHe consumption and cooling effect of other cryogenic liquids. Therefore, in order to better obtain the cooling effect of LH_2 , it is necessary to further understand the simultaneous heat and mass transfer processes. In the present paper, a theoretical model is developed to reveal the simultaneous heat and mass transfer mechanism. On this basis, a thermodynamic model considering effects of static pressure and thermophysical properties is proposed to describe the cooling process. The calculating results are then compared with Schmidt's and Cho's experimental data for verification. Meanwhile, effects of several factors, including liquid evaporation, helium solubility, tank height, ullage pressure, GHe injection rate, GHe injection temperature, environmental heat flux, and various cryogenic liquids are discussed to optimize the cooling method of GHe injection.

2 Mathematical method

2.1 Simultaneous heat and mass transfer model

When helium bubbles are injected into LH_2 , the heat and mass transfers occur simultaneously. To visually describe these processes, imaginary surfaces on both sides of the real interface are denoted based on the two-film theory [28], as shown in Fig. 3. The real interface, the vapor film interface, and the liquid film interface are indicated by i , s , and u , respectively.

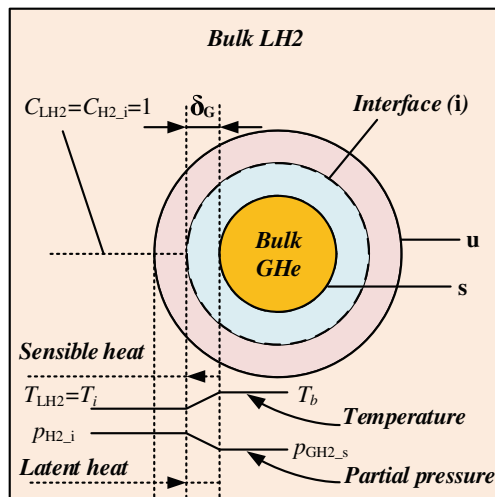


Fig. 3 Simultaneous heat and mass transfer model

Ignoring a small amount of helium dissolved in LH₂, the hydrogen concentration at the 'u' surface $C_{LH_2,u}$ and the 'i' interface $C_{LH_2,i}$ is 1. For heat transfer behaviors, the liquid-side thermal resistance and interfacial thermal resistance can be ignored, and the resistance is mainly concentrated on the gas side. As seen in Fig. 3, the temperature difference and concentration difference between the 'i' interface and 's' surface cause the heat and mass transfer simultaneously in the vapor film. The vapor film thickness is denoted by δ_G . The solubility of GHe-LH₂ is discussed in detail in Section 2.2. Except for extreme circumstances, the mixture of hydrogen vapor and GHe at the 'i' interface must be in thermodynamic equilibrium. The hydrogen concentration at the 's' surface is determined by the partial pressure of bulk hydrogen $p_{GH_2,s}$ in the bubble. The temperature of the 's' surface is expressed by the bulk vapor temperature which is assumed to distribute uniformly. The temperatures of the 'u' surface and the 'i' interface are denoted by the bulk LH₂ temperature which is also assumed to be uniform.

If a hot helium bubble is injected into LH₂, e.g., the 200 K helium bubble is injected into the 16 K LH₂, the heat transfer between the hot helium bubble and LH₂ may experience forced convection boiling (film boiling, transition boiling and nucleate boiling) and forced convection of single phase successively. Because the boiling heat transfer mechanism on forced convection is very complex, and the cooling effect of hot helium bubbling discussed in Section 4.2 is poorer than cool helium gas, the forced convection of single phase is only considered to describe the unsteady heat transfer process, e.g., 20 K helium bubbles are injected into 16 K LH₂.

The simultaneous heat and mass transfer for the single bubble are regarded as unsteady problem, and conservation equations for mass and energy are listed as follows:

$$qA_b + Q_c = -(m_{GH_2}c_{pGH_2} + m_{GHe}c_{pGHe}) \frac{dT_b}{d\tau} \quad (1)$$

$$kA_b \left(\frac{p_{GH_2,i}}{R_{GH_2}T_{LH_2}} - \frac{p_{GH_2,s}}{R_{GH_2}T_b} \right) = \frac{dm_{GH_2}}{d\tau} \quad (2)$$

The total sensible heat flux is expressed as

$$q = h \left(-\frac{dT}{d\delta} \right) \delta_G + \dot{m}_{GH_2,diff} c_{pGH_2} (T - T_{LH_2}) \quad (3)$$

Rearranging and integrating give

$$\int_{T_{LH_2}}^{T_b} \frac{dT}{q - \dot{m}_{GH_2,diff} c_{pGH_2} (T - T_{LH_2})} = \frac{1}{h\delta_G} \int_{\delta_G}^0 d\delta \quad (4)$$

$$q = \frac{\dot{m}_{GH_2,diff} c_{pGH_2}}{1 - e^{-\dot{m}_{GH_2,diff} c_{pGH_2}/h}} (T_b - T_{LH_2})$$

The latent heat flux is given as

$$Q_c = \dot{m}_{GH_2,diff} [H_{GH_2}(T_{LH_2}, \varphi = 1) - H_{LH_2}(T_{LH_2}, \varphi = 0)] \quad (5)$$

The heat transfer between the bubble and LH₂ is treated as a problem of external flow over spheres. The correlation of Ranz and Marshall [29] is used, as follows:

$$Nu = 2 + 0.6Re^{1/2}Pr^{1/3} \quad (6)$$

The range of Reynolds number in Eq. (6) is from 0 to 200, and the typical operation conditions in this paper are in this range.

Because the convection heat and mass transfer processes are governed by dimensionless equations of the same form, the processes are analogous [30]. The analogy may also be used to directly relate the two convection coefficients, as follows:

$$\frac{h}{k} = \frac{\lambda_m}{D_{GH_2GHe}Le^{1/3}} \quad (7)$$

According to the Fuller correlation [31], the diffusion coefficient is expressed, and the key parameters are given as shown in Table 1, as follows:

$$D_{GH_2GHe} = 0.01 \times \frac{T_f^{1.75} (1/M_{H_2} + 1/M_{He})^{1/2}}{p_t [\omega_{H_2}^{1/3} + \omega_{He}^{1/3}]^2} \quad (8)$$

Because of the gasification of LH₂ into the bubble, the partial pressure of hydrogen vapor in the bubble gently

Table 1 Diffusion volumes for simple molecules

Species	Diffusion volumes	M	Species	Diffusion volumes	M
H ₂	7.07	2.016	N ₂	17.9	28.00
He	2.88	4.003	O ₂	16.6	32.00

increases, and is expressed according to the ideal gas equation:

$$p_{GH2s} = \frac{m_{GH2} R_{gH2} T_b}{V_b} \quad (9)$$

According to the Raoult's law [30], the partial pressure of hydrogen to reach phase equilibrium is determined as

$$p_{GH2i} = x_{LH2} p_{H2sat} \quad (10)$$

Because GHe is insoluble in LH₂, x_{LH2} is equal to 1. Then Eq. (10) can be simplified to $p_{GH2i} = p_{H2sat}$.

Based on the Dalton's law [30], the relation between the total pressure and the partial pressure is written as:

$$p_t = p_{GH2i} + p_{GHei} = p_{GH2s} + p_{GHe} \quad (11)$$

In the processes of simultaneous heat and mass transfer, the bubble volume is changed. The calculating equation is written, as follows:

$$V_b = \frac{(n_{GH2} + n_{GHe}) R_g T_b}{p_t} \quad (12)$$

$$d_b = 2 \times \left(\frac{3V_b}{4\pi} \right)^{1/3} \quad (13)$$

$$A_b = 4\pi(d_b/2)^2 \quad (14)$$

$$m_{GH2} = \int_0^\tau \dot{m}_{GH2diff} d\tau \quad (15)$$

$$n_{GH2} = \frac{m_{GH2}}{M_{H2}} \quad (16)$$

When GHe is blown steadily through a submerged orifice into LH₂, the GHe emerges as bubbles which are formed periodically over a wide range of flow rates. However, the process of bubble formation is controlled by many influencing parameters, such as surface tension, viscosity of liquid and gas, density of liquid and gas, orifice diameter, contact angle, injection rate of gas, velocity of sound in gas, and so forth. The main forces acting on the detachment and movement of the bubble are buoyancy force, gas momentum force, pressure force, drag force, inertial force, and surface tension force. They reach a balance around the spherical bubble and would detach the bubble when the neck between the orifice and bubble is broken off.

In the process of industrial operations, not all the forces work simultaneously, and some of them could be ignored in the analysis. Therefore, the different flow regimes are formed, and have significant effect on the performance of bubbles. Three types of flow regimes are commonly observed in bubble columns, including the homogeneous (bubbly flow)

regime, the heterogeneous (churn-turbulent) regime, and slug flow regime [32–37].

In the present work, since the rates of collision and agglomeration of bubbles are lower at homogeneous regime compared with the heterogeneous regime, the bubbly flow regime is considered. Davidson model [34] neglected the momentum of helium gas is recommended to obtain the final bubble size at constant flow rate:

$$V_b = \left(\frac{4\pi}{3} \right)^{1/4} \left(\frac{15v_{LH2} \dot{V}}{2g} \right)^{3/4} \quad (17)$$

After a bubble detaches from the orifice, it rapidly reaches a terminal velocity u under the influence of gravity. For bubbles with diameter greater than 2 mm, Mandelson [38] proposed the bubble terminal velocity according to wave theory, since it unifies the existing correlations for both the surface tension and buoyancy dominated regimes, as follows:

$$u = \left(\frac{\sigma}{\frac{d_b}{2} \rho_{LH2}} + g \frac{d_b}{2} \right)^{1/2} \quad (18)$$

2.2 Solubility of GHe in LH₂

Zimmerli's correlation [39] to calculate the helium solubility in LH₂ in the range of 14 K < T < 29 K and 0 < p_t < 5 MPa is used as follows:

$$x_{He} = p^* \cdot \exp[a(T) + b(T) \cdot p^*] \quad (19)$$

where x_{He} is the mole fraction solubility of helium in LH₂, p^* is the dimensionless pressure defined as $p^* = (p_t - p_{LH2sat})/p_{REF}$, where p_{REF} is the reference pressure defined as equal to 1 MPa, and $a(T)$ and $b(T)$ are empirically determined as

$$a(T) = -9.338 + \frac{0.3627}{K} T - \frac{0.00516}{K^2} T^2$$

$$b(T) = -0.3124 - \frac{0.0006}{K} T + \frac{0.0004}{K^2} T^2$$

The percent relative standard deviation of the data from the fitted value over the range of data is $\sigma_R = 7.5\%$ ($N = 245$).

Figure 4 shows the change of the mole fraction solubility of helium in LH₂ with temperature at different pressures. As seen from this figure, the equilibrium helium solubility goes up as LH₂ temperature and total pressure increase. Nevertheless, it is found that the total pressure has remarkable effect on the cooling capacity in Section 4.3, so the total pressure cannot be too high, which keeps at one atmosphere pressure in practical applications. Consequently, the helium solubility in LH₂ is negligible.

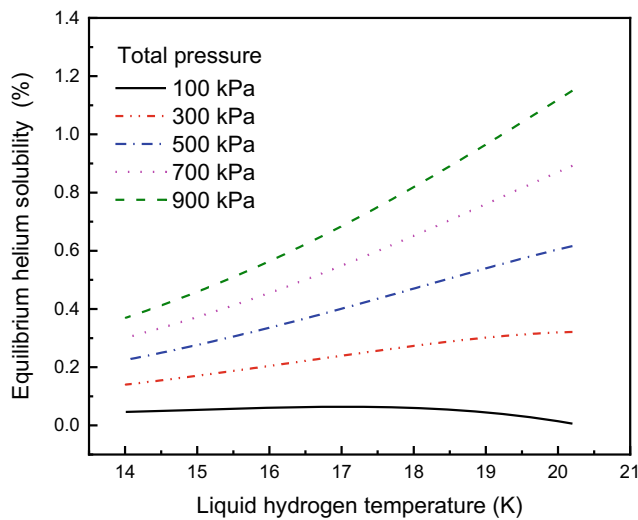


Fig. 4 Mole fraction solubility of helium in LH₂ at different pressures

In the absence of any mixing mechanism, the diffusion of helium molecules in LH₂ is governed by a diffusion equation. In contrast to diffusion of hydrogen molecules in GHe, the resistance of helium molecules diffusing into LH₂ is mainly concentrated on the liquid side. This can be considered as a conduction problem of one-dimensional flat. The liquid film thickness is denoted by δ_u . The equation is expressed as

$$\frac{\partial^2 C_{He}}{\partial x^2} = \frac{1}{D_{LH_2GHe}} \frac{\partial C_{He}}{\partial \tau} \quad (20)$$

Assuming an initial uniform concentration,

$$C_{He}(x, 0) = 0 (0 \leq x \leq \delta_u)$$

The equilibrium concentration of helium at the interface 'i' is

$$C_{He}(0, \tau) = C_{Hei}$$

Hence the concentration distribution in liquid film can be expressed as

$$C_{He}(\delta_u, \tau) = C_{Hei} - C_{Hei} \cdot \text{erf} \left(\frac{\delta_u}{2\sqrt{D_{LH_2GHe}\tau}} \right) \quad (21)$$

According to Eq. (21), the concentration at location δ_u attains half of its equilibrium value when $\tau \approx \delta_u^2 / D_{LH_2GHe}$, which can be considered as a characteristic timescale for one-dimensional diffusion.

The Stokes-Einstein equation [31] is used to calculate the mass diffusion coefficient for helium diffusing into LH₂, as follows:

$$D_{LH_2GHe} = \frac{k_B T_{LH_2}}{6\pi\mu_{LH_2}R_0} \quad (22)$$

where k_B is Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/K, μ_{LH_2} is the dynamic viscosity coefficient of LH₂, and R_0 is the helium radius assumed to be half of the collision diameter in the gas, $R_0 = 1.28 \times 10^{-10}$ m.

At the normal boiling point temperature of LH₂, namely 20.4 K, the helium mass diffusion coefficient in LH₂ is calculated by Eq. (22) to be $D_{LH_2GHe} = 8.78 \times 10^{-9}$ m²/s. The characteristic timescale for the helium diffusing 0.5 mm deep into LH₂ is about 28.5 s, while the contact time between the bubble and liquid is very transient due to the movement of the single bubble by the buoyancy force. As a result, the equilibrium solubility has not been reached in industrial applications.

From the above results, it is found that the helium mole fraction solubility is far less than 0.1% in typical cooling system, and the time to reach equilibrium is very long. Consequently, it can be assumed that the mass transfer of GHe diffusing into LH₂ can be ignored.

2.3 Cooling capacity of single bubble

For single bubble, the heat and mass transfers are regarded as steady state problem, and the conservation equation of energy is given as follows:

$$E = m_{GH_2} \gamma_{LH_2} - m_{GHe} (H_{GHeout} - H_{GHein}) \quad (23)$$

The state parameters of each component to reach phase equilibrium can be expressed in state equations of real gases, respectively:

$$p_{GH_2} V_b = Z_{GH_2} m_{GH_2} R_g T_b / M_{H_2} \quad (24)$$

$$p_{GHe} V_b = Z_{GHe} m_{GHe} R_g T_b / M_{He} \quad (25)$$

The hydrogen mass in the mixture bubble is simplified as

$$m_{GH_2} = m_{GHe} \frac{Z_{GHe} p_{GH_2} M_{H_2}}{Z_{GH_2} p_{GHe} M_{He}} \quad (26)$$

The partial pressure of hydrogen to reach phase equilibrium is determined with Eq. (10), i.e., $p_{GH_2} = p_{H_2, sat}$. Based on Dalton's law, the relation between the total pressure inside the bubble and the partial pressure is written as

$$p_{inside} = p_{H_2, sat} + p_{GHe} \quad (27)$$

Based on the Young-Laplace equation [40], interfacial energy of the helium gas bubble is expressed:

$$p_{inside} - p_{outside} = \frac{4\sigma}{d_b} \quad (28)$$

The static pressure energy is given using hydrostatics:

$$p_{outside} = p_o + \rho_{LH_2} g L \quad (29)$$

The simplification equation from Eq. (27) to Eq. (29) is derived as follows:

$$p_{GHe} = p_o + \rho_{LH_2} g L + \frac{4\sigma}{d_b} - p_{H_2 sat} \quad (30)$$

The initial helium mass in the bubble is expressed in state equation of real gases

$$m_{GHe} = \frac{p_{inside} V_b}{Z_{GHe} R_{GHe} T_b} \quad (31)$$

2.4 Thermodynamic model

Since the cooling process of pure helium bubbles injected into LH_2 is very complex, it is necessary for reasonable simplification. Based on the analysis of simultaneous heat and mass transfer of single bubble, many hypothesis conditions are given as follows:

- (a) As duration time of helium bubbles in LH_2 is very short for the entire subcooling period, and bubbling is continuous for a long period of time, a fictitious gas phase region is expressed by the dotted line, as shown in Fig. 5.

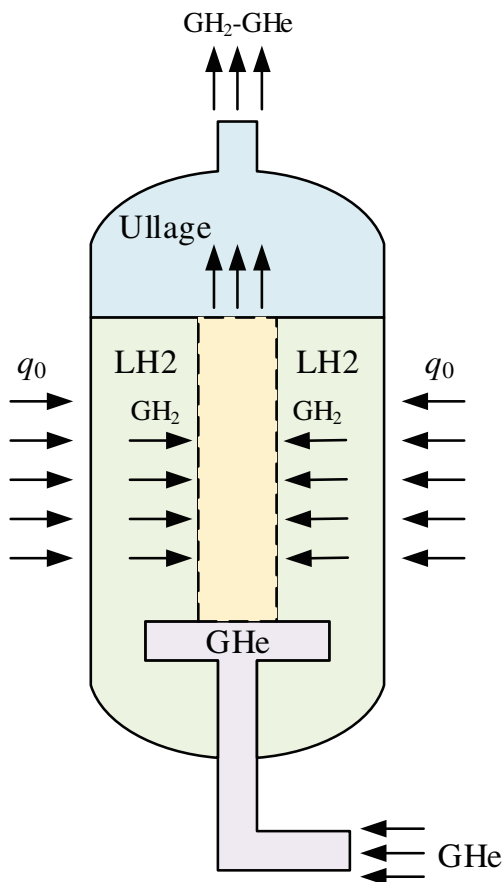


Fig. 5 Thermodynamic model

- (b) The dynamics of gas injection process are based on the assumption of instantaneous phase equilibrium in the cooling system, i.e., heat and mass transfer processes of the pure helium bubbles instantaneously reach phase equilibrium.
- (c) There is no thermal stratification in LH_2 .
- (d) Helium gas is insoluble in LH_2 .
- (e) The interfacial energy of bubbles is negligible.
- (f) The gas phase is considered as a mixture of ideal gases.
- (g) The gas injection rate is constant.

With reference to the thermodynamic model shown in Fig. 5, the continuity equation for GHe is given as

$$\frac{dm_{GHe}}{d\tau} = \dot{m}_{GHein} - \dot{m}_{GHeout} - \dot{m}_{GHeS} \quad (32)$$

Based on assumption (a), the change of mass flow rate for GHe in the fictitious gas phase volume is expressed as

$$\frac{dm_{GHe}}{d\tau} = 0 \quad (33)$$

Based on assumption (d), the GHe mass dissolved into LH_2 is written as

$$\dot{m}_{GHeS} = 0 \quad (34)$$

And eq. (32) can be reduced to

$$\dot{m}_{GHe} = \dot{m}_{GHein} = \dot{m}_{GHeout} \quad (35)$$

The continuity equation for LH_2 is given as

$$\frac{dm_{LH_2}}{d\tau} = \dot{m}_{GH_2} \quad (36)$$

The state parameters of each component to reach phase equilibrium in the fictitious gas phase volume can be expressed with Eqs. (24) and (25).

The mole fraction ratio is defined as

$$X = \frac{n_{GHe}}{n_{GH_2}} = \frac{Z_{GH_2} p_{GHe}}{Z_{GHe} p_{GH_2}} \quad (37)$$

Differentiating Eq. (37), we get

$$\frac{\dot{n}_{GH_2}}{\dot{n}_{GHe}} = \frac{1}{X} - \frac{\dot{X}}{X^2} \frac{n_{GH_2}}{\dot{n}_{GHe}} \quad (38)$$

Based on assumption (b), $\dot{X} = 0$, Eq. (38) is simplified to

$$X = \frac{\dot{n}_{GHe}}{\dot{n}_{GH2}} = \frac{\dot{m}_{GHe} R_{gGHe}}{\dot{m}_{GH2} R_{gGH2}} \quad (39)$$

Substituting Eq. (37) into Eq. (39), a simplifying Eq. (40) is obtained as

$$\frac{\dot{m}}{\dot{m}_{GH2}} = \frac{\dot{m}_{GHe}}{\dot{m}_{GH2}} \frac{Z_{GHe} p_{GH2}}{Z_{GH2} p_{GHe}} \frac{R_{gGHe}}{R_{gGH2}} \quad (40)$$

In gas phase volume, equilibrium partial pressure of hydrogen vapor is given with Eq. (10). Based on assumption (e), Eq. (30) is simplified to

$$p_{GHe} = p_o + \rho_{LH2} g L_z - p_{H2sat} \quad (41)$$

The energy equation for LH₂ is given as

$$\frac{d(m_{LH2} H_{LH2})}{d\tau} = \dot{m}_{GHe} (H_{GHeout} - H_{GHein}) - \dot{m}_{GH2} \gamma_{H2} + Q_e \quad (42)$$

Equation (42) is simplified to

$$\begin{aligned} m_{LH2} \frac{dH_{LH2}}{d\tau} &= \dot{m}_{GHe} (H_{GHeout} - H_{GHein}) - \dot{m}_{GH2} \gamma_{H2} + Q_e - H_{LH2} \frac{dm_{LH2}}{d\tau} \\ &= \dot{m}_{GHe} (H_{GHeout} - H_{GHein}) - \dot{m}_{GH2} \gamma_{H2} + Q_e - H_{LH2} \frac{dm_{LH2}}{d\tau} \end{aligned} \quad (43)$$

Since the pressure has little effect on the liquid enthalpy, the liquid enthalpy is given, as follows:

$$H = c_{pLH2} T \quad (44)$$

Substituting Eqs. (36) and (44) into Eq. (43), we get

$$\frac{dT}{d\tau} = \frac{\dot{m}_{GHe} (H_{GHeout} - H_{GHein}) - \dot{m}_{GH2} (\gamma_{H2} + c_{pLH2} T_{LH2}) + Q_e}{c_{pLH2} m_{LH2}} \quad (45)$$

The formulation given by Eq. (45) is subject to the following initial conditions:

$$\begin{aligned} \tau = 0, \quad m_{LH2} &= m_{LH20}, \quad T_{LH2} = T_{LH20}, \quad p_o = p_{o0}, \quad L = L_0, \\ \dot{m}_{GHe} &= \dot{m}_{GHe0}, \quad Q_e = Q_{e0}, \quad \text{and } T_b = T_{b0}. \end{aligned}$$

2.5 Solution strategy

Since many parameters including mass, volume, thermalphysical properties, and so forth may change with the decrease of LH₂ temperature in the cooling process, this process cannot be accurately described by an analytical

solution. Therefore, an infinitesimal element method is used to solve this problem. Discrete equations are as follows:

$$p_{H2sat}^i = f(T^i, \varphi = 1)$$

$$\rho_{LH2}^i = f(p_t, T^i)$$

$$p_{GHe}^i = p_o + \rho_{LH2}^i g L_z - p_{H2sat}^i$$

$$Z_{GHe}^i = f(T^i, p_{GHe}^i) \quad Z_{GH2}^i = f(T^i, p_{GH2}^i)$$

$$\frac{\dot{m}}{\dot{m}_{GH2}} = \frac{\dot{m}_{GHe}}{\dot{m}_{GH2}} \frac{Z_{GHe}^i p_{GH2}^i}{Z_{GH2}^i p_{GHe}^i} \frac{R_{gGHe}}{R_{gGH2}}$$

$$H_{LH2}^i = f(T^i, \varphi = 0) \quad H_{GH2}^i = f(T^i, \varphi = 1)$$

$$\gamma_{H2}^i = H_{GH2}^i - H_{LH2}^i$$

$$c_{pLH2}^i = f(T^i, \varphi = 0)$$

$$H_{GHein}^i = f(T_b, p_t) \quad H_{GHeout}^i = f(T^i, p_{GHe}^i)$$

$$m_{LH2}^{i+1} = m_{LH2}^i - \dot{m}_{GH2}^i d\tau$$

$$T^{i+1} - T^i = \frac{\dot{m}_{GHe}^i (H_{GHein}^i - H_{GHeout}^i) - \dot{m}_{GH2}^i [1 \cdot (\gamma_{H2}^i + c_{pLH2}^i T^i)] + Q_e}{c_{pLH2}^i m_{LH2}^i} d\tau$$

The above algorithm is run by MATLAB programs and the relevant physical parameters are obtained by calling NIST data [41]. The simplified flowchart of the algorithm is shown in Fig. 6.

3 Verification of thermodynamic model

To verify accuracy of the proposed thermodynamic model, the calculated results are compared with associated experimental data [1, 11]. For Schmidt's experiment, the cryostat was of 72 L. The central test column was constructed of a stainless steel pipe with 152.4 mm in diameter and 3.505 m in height. A thermal buffer plug made of polystyrene foam was installed at the top of the pipe. Prior to experiment, the experimental column was filled with LH₂ until the condition of overflow was observed at the vent orifice, and the filled volume of LH₂ was 63.9 L. The GHe injection rate was 4.7 L/s, ullage pressure was 90 kPa, and environmental heat flux was 4.6 W. The injection temperature of helium gas was 200 K, 77 K, and 20 K, respectively.

Figure 7 presents the calculated results compared with experimental data at different injection temperatures. It is found that the calculated values are in good agreement with Schmidt's experimental data. At 20 K, 77 K, and 200 K helium injection temperature, the maximum relative error is 2.32%, 1.2%, 1.3%, respectively.

The Cho's experimental setup was a test tank of 90 L, 2.5 m height by 0.23 m diameter [11]. The vacuum and multi-layer insulation were used in the tank. The tank pressure could be

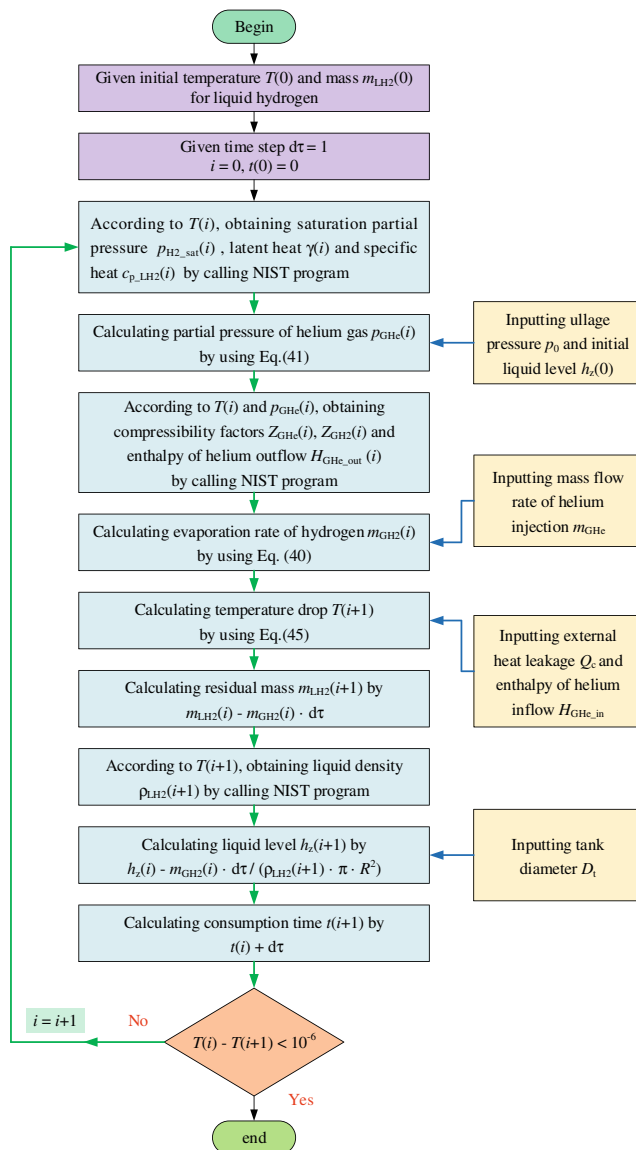


Fig. 6 Simplified flowchart of the algorithm

controlled by atmospheric venting and a back pressure regulator installed at the vent port. At different helium injection flow rates, the experimental conditions were shown in Table 2.

Figure 8 shows the compared results of LO₂ under different helium injection rates. As shown in this figure, the calculated values are consistent with Cho's experimental data. At 55.2 L/min, 90.6 L/min, 134.5 L/min helium injection flow rate, the maximum relative error is 0.2%, 0.3%, 0.5%, respectively. The relative error has a rising trend with the increase of helium injection flow rate. It is speculated that this is due to inaccurate measurement of ullage pressure. Since venting capacity does not improve with the increase of injection gas flow rate, and pressure loss of pipelines also increase resulting in the increase of ullage pressure, the calculated results at the initial ullage pressure have a slight downward trend.

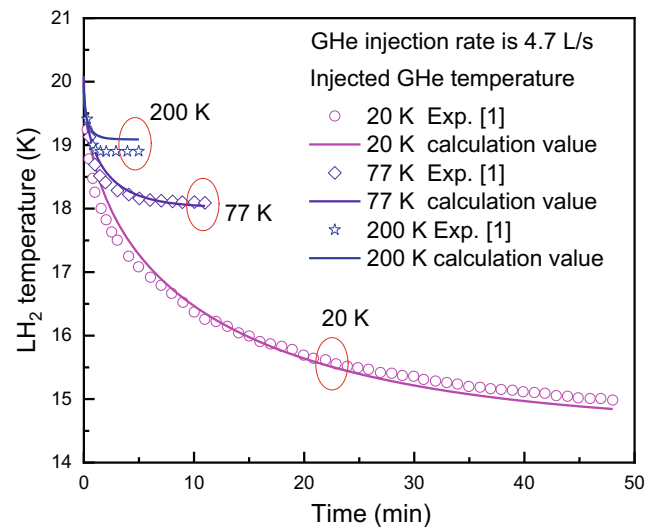


Fig. 7 LH₂ temperature profiles with time at different injected GHe temperatures

4 Results and discussion

4.1 Simultaneous heat and mass transfer processes

As shown in Table 3, the initial conditions are performed on simultaneous heat and mass transfer between single helium bubble and LH₂. Several factors, including bubble temperature, evaporation mass of hydrogen diffusing into the bubble, concentration of hydrogen in the bubble, and bubble diameter are analyzed with Eqs. (1)–(16).

Figure 9 depicts the simultaneous heat and mass transfer processes of 20 K pure helium bubbles injected into 16 K LH₂. As seen in Fig. 9a, the bubble temperature drops to 16 K during about 0.18 s. In other words, it needs only about 0.18 s for the bubble to reach its thermal equilibrium state. Figure 9b indicates that the partial pressures of hydrogen and helium reach concentration equilibrium after about 4.5 s, and the mass transfer occurs mainly in the early stage (1.5 s for this case), since the capacity of mass transfer gradually weakens with the decreasing hydrogen concentration difference in the bubble. It is found that the mass transfer time is slower 1 order of magnitude than the heat transfer time at least.

Table 2 Test conditions [11]

Test conditions	Case 1	Case 2	Case 3
Helium injection flow rate (L/min) ($\dot{V}_{GHe}/V_{LOx}, \text{min}^{-1}$)	55.2 (0.85)	90.6 (1.39)	134.5 (2.07)
Filled LO ₂ volume (L)	65.0	65.0	65.1
Ullage pressure (MPa)	0.11	0.11	0.11
Initial LO ₂ temperature (K)	91.0	91.0	91.0
Ambient heating rate (W)	261.0	261.0	261.0
Helium injection temperature (K)	91.0	91.0	91.0

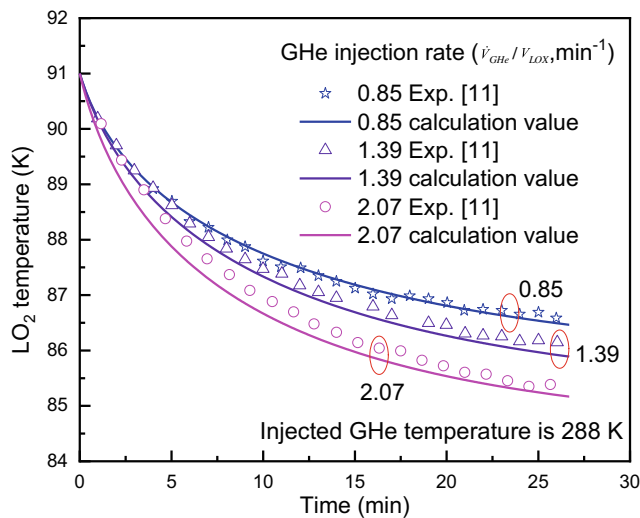
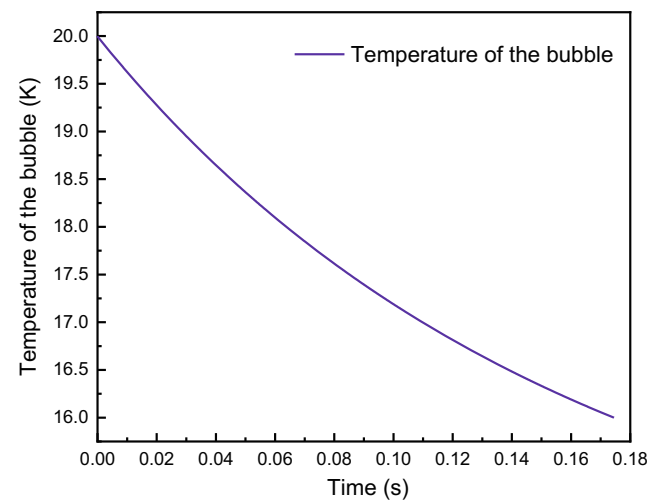


Fig. 8 LO₂ temperature profiles at different helium injection flow rates

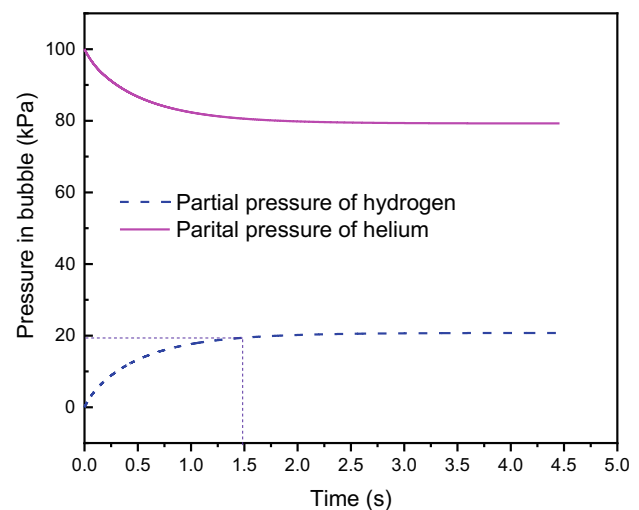
Different influence factors on the simultaneous heat and mass transfer processes are studied and shown in Fig. 10. It is clearly seen that the mass transfer process of 20 K helium bubbles injected into 20 K LH₂ to reach concentration equilibrium needs about 50 s, but mainly concentrates in the first 10 s in Fig. 10a, while the heat transfer process is very transient in Fig. 10b. That is to say, the mass transfer process is not instantaneously completed, especially under the boiling point state of LH₂, but the heat transfer process can be ignored. Consequently, in the design of a cooling device, phase equilibrium of each bubble in the mass transfer process should be guaranteed. If the bubbles flowing out from LH₂ do not achieve phase equilibrium, GHe will waste more. In practical applications, in order to assure reaching phase equilibrium of each bubble, lengthening the device height should be considered to increase the moving time of the bubble in LH₂.

Figure 10c and d show the simultaneous heat and mass transfer processes at different GHe injection rates. It can be observed that the time to reach phase equilibrium

(concentration equilibrium and thermal equilibrium) increases with the increase of GHe injection rate. Since the injected bubble diameter goes up with the increasing GHe injection rate, the time to reach phase equilibrium also gradually extends. Figure 10e and f display the simultaneous heat and mass transfer processes at different helium injection temperatures. It is clear that the time to reach concentration equilibrium slightly decreases with the increase of injected GHe temperature. It indicates that the injected GHe temperature has slight effect on the concentration equilibrium. For heat transfer process, the time to reach thermal equilibrium goes up with the increase of injected GHe temperature. Figure 10g and h reflect that the time to reach phase equilibrium increases as the total pressure goes up. Since the mass diffusion coefficient declines with the increasing total pressure, that is to say, the



(a) temperature profile in the bubble



(b) partial pressure profiles in the bubble

Fig. 9 Simultaneous heat and mass transfer processes in the bubble

Table 3 Initial conditions for heat and mass transfer analysis

Initial conditions	Value	Note
LH ₂ temperature (K)	16.0	
Total pressure (kPa)	100.0	
Helium injection rate (L/s)	4.7	
Injection helium temperature (K)	20.0	
Diameter of the bubble (mm)	7.8	Calculated with Eq. (13)
Volume of the bubble (mm ³)	245.89	Calculated with Eq. (12)
Area of the bubble (mm ²)	189.81	Calculated with Eq. (14)
Heat transfer coefficient W/m ² K	51.21	Calculated with Eq. (6)
Mass transfer coefficient cm/s	0.2965	Calculated with Eq. (7)
Mass diffusivity (mm ²)	1.2163	Calculated with Eq. (8)

capacity of mass transfer weakens, resulting in the increase of the time to reach concentration equilibrium.

4.2 Influence factors of cooling capacity

The cooling capacity of single helium bubble injected into LH₂ is discussed at different influence factors. The initial conditions are given, as shown in Table 4.

Figure 11a illustrates the cooling capacity of single helium bubble injected into LH₂ at different injected GHe temperatures. From this figure it can be seen that the cooling capacity of single helium bubble decreases with the decreasing LH₂ temperature, and increase with the decrease of injected GHe temperature. Because the corresponding saturation pressure goes down with the decrease of LH₂ temperature, resulting in the decrease of concentration difference, the cooling effect becomes poorer. The higher the helium temperature is, the more heat to bring into the cooling system is. At this time, a part of the refrigeration capacity will be offset by the sensible heat,

resulting in the decrease of the cooling effect. As a result, it is recommended that a set of pre-cooling device in the cooling system should be added to obtain better cooling effect during LH₂ cooling. Figure 11b shows the effects of total pressure and GHe injection rate on cooling capacity of single bubble. It is clearly observed that the cooling capacity of single helium bubble goes down with the increase of total pressure, and goes up with the increase of GHe injection rate. Since the GHe injection rate is related to the bubble diameter, i.e., the greater the bubble diameter is, the more the helium mass in the bubble is. Meanwhile, from Eq. 18, it can be seen that the greater the bubble diameter, the faster the rising velocity of the bubble in LH₂, so the bubble cannot reach the phase equilibrium after passing through LH₂. Moreover, it is found that higher total pressure is unbeneficial for reaching the phase equilibrium.

To further study the influences of the total pressure including the ullage pressure, the static pressure of liquid and interfacial energy of the bubble, pressure ratios are defined, respectively, i.e., $4\sigma d_b^{-1}/p_o$ and $\rho_{LH_2}gL/p_o$. Figure 11c indicates the

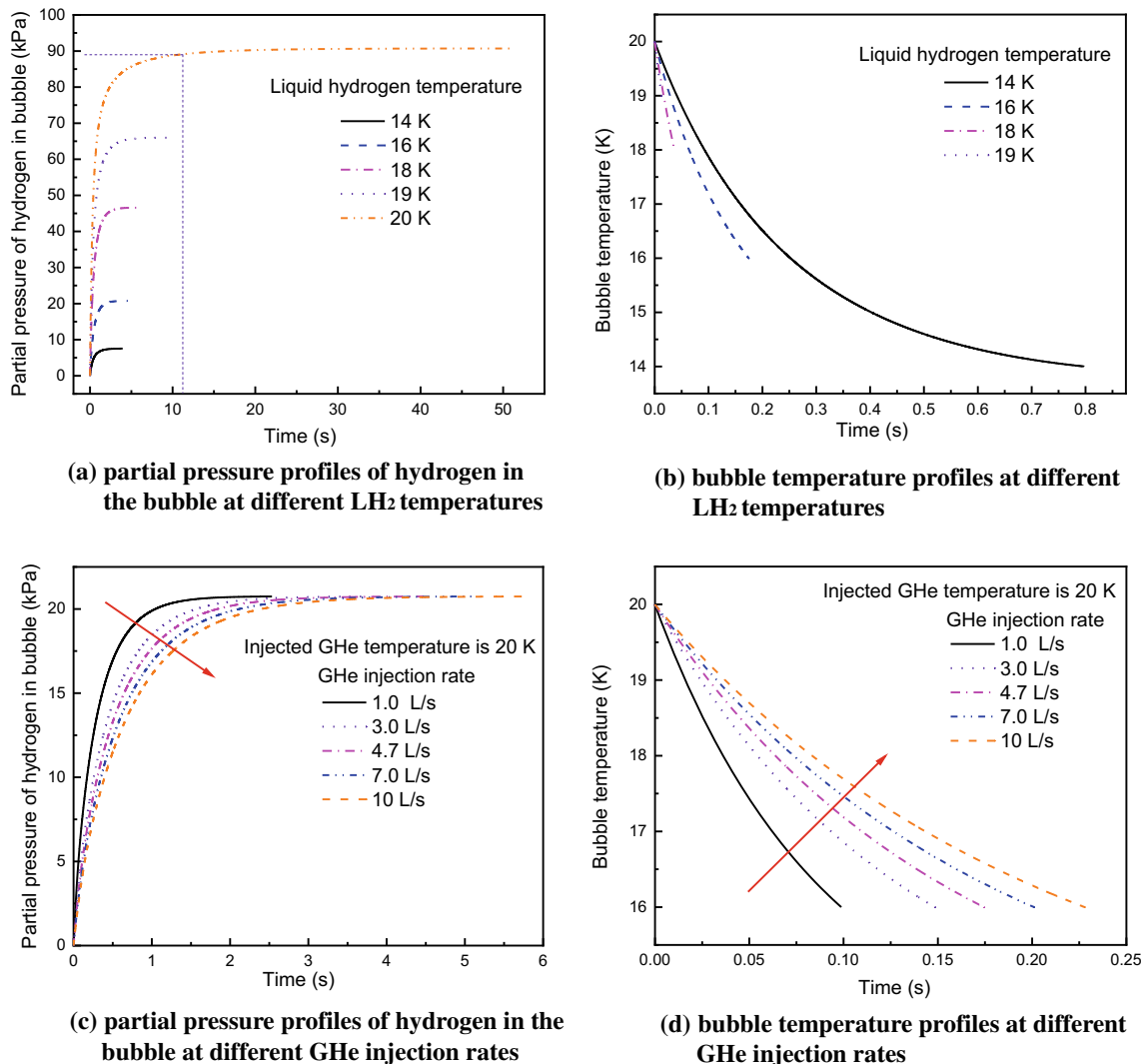
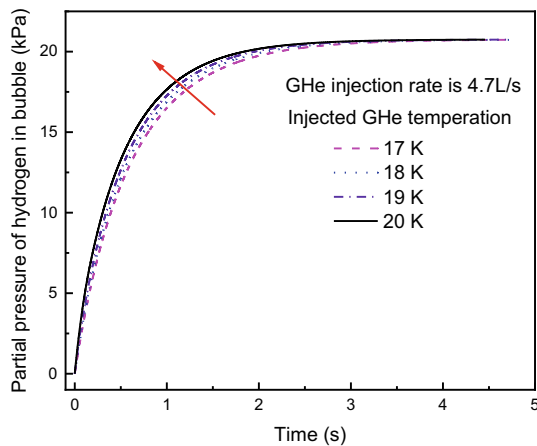
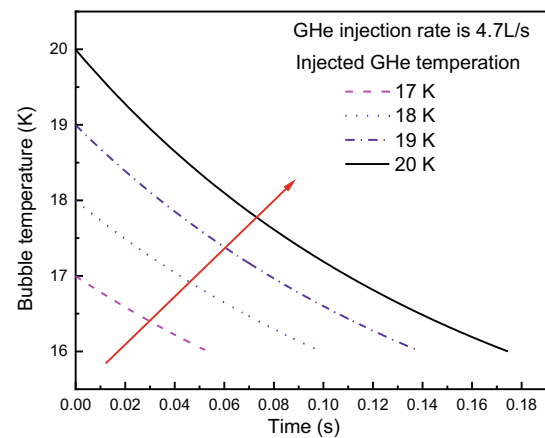


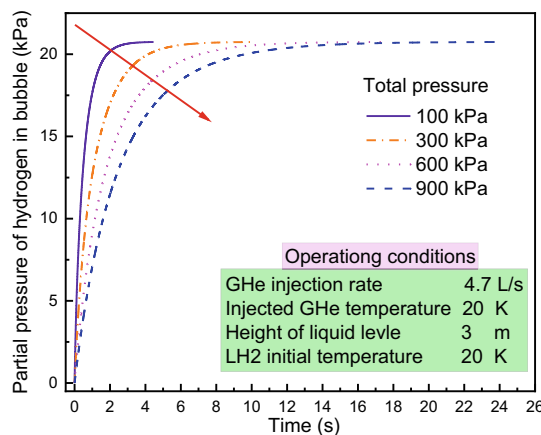
Fig. 10 Influence law of different factors on simultaneous heat and mass transfer processes



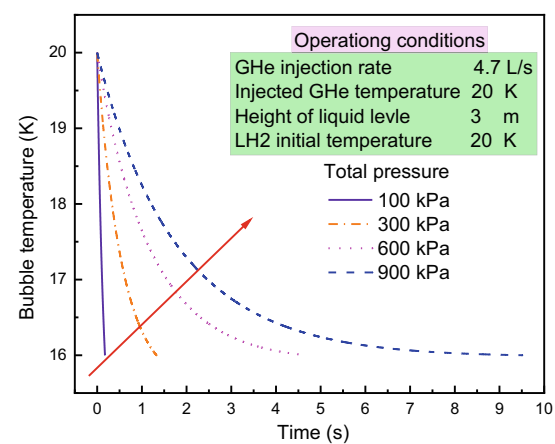
(e) partial pressure profiles of hydrogen in bubble at different GHe injection temperatures



(f) bubble temperature profiles at different GHe injection temperatures



(g) partial pressure profiles of hydrogen in the bubble at different total pressures



(h) bubble temperature profiles at different total pressures

Fig. 10 (continued)

pressure ratio profiles with varying bubble diameter at different LH_2 temperatures. From this figure, it is clearly seen that the proportion of interfacial energy goes up with the decreasing bubble diameter, and declines with the increasing LH_2 temperature. It is found that the pressure ratio of interfacial energy only reaches to 1.3% at 0.01 mm diameter and 14 K temperature which is worse working conditions. In practical applications, in order to obtain better cooling effect, the bubble diameter is generally greater than 3 mm. Consequently, the proportion of interfacial energy of the bubble is very small, only about $10^{-3}\%$, and its effect can be ignored. Figure 11d shows the pressure ratio profiles with varying height of liquid

level at different LH_2 temperatures. From this figure it is observed that the proportion of static pressure increases with the enhancement of liquid level height, and decreases with the increase of LH_2 temperature. It is found that the proportion of static pressure of liquid level will reach to 7.5% at 10 m height and 14 K temperature. In the design of cooling system, it is necessary to determine a certain height of tank to ensure the bubbles have enough time to reach phase equilibrium, so the static pressure of liquid level cannot be ignored in the calculating total pressure.

Consequently, the total working pressure has significant effect on the design of cooling system, which an optimum value exists.

Table 4 Initial conditions for cooling capacity of single bubble

Parameter	Value	Parameter	Value
GHe injection rate (L/s)	4.7	Ullage pressure (kPa)	101.325
Injected GHe temperature (K)	20.0	Height of liquid level (m)	3.0
LH_2 temperature (K)	20.0		

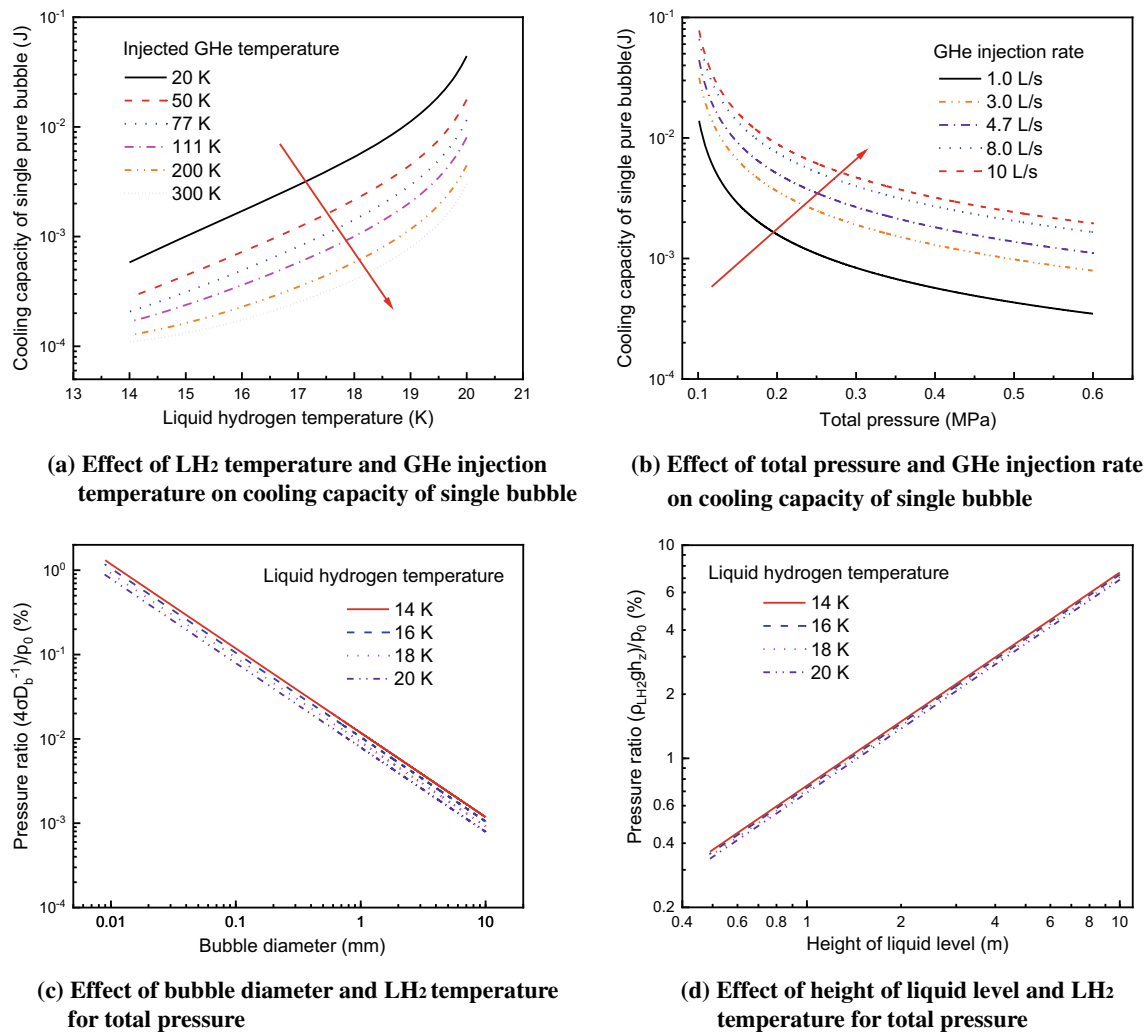


Fig. 11 Influence law of different factors on cooling capacity of single helium bubble

For the ullage pressure, this operating condition should be slightly greater than the ambient pressure, and be maintained at a constant value. The external impurities will infiltrate to pollute the liquid hydrogen and cause safety accidents if the pressure is too low. However, the cooling effect will be affected if the pressure is too high. As a result, in the actual operation process, in order to discharge the ullage pressure in time, it is possible to

increase the number of discharge orifice, the diameter of discharge orifice, or a mechanical pump for pumping out directly.

Table 5 Input conditions on thermodynamic model analysis

Parameters	Value	Parameters	Value
Diameter of tank (m)	1.0	Duration time (min)	180.0
Height of tank (m)	3.0	GHe injection rate (N m ³ /min)	6.0 m ³ /min
LH ₂ temperature (K)	20.4	Environmental heat flux (W)	300.0
Injected GHe temperature (K)	21.0	Ullage pressure (kPa)	101.325

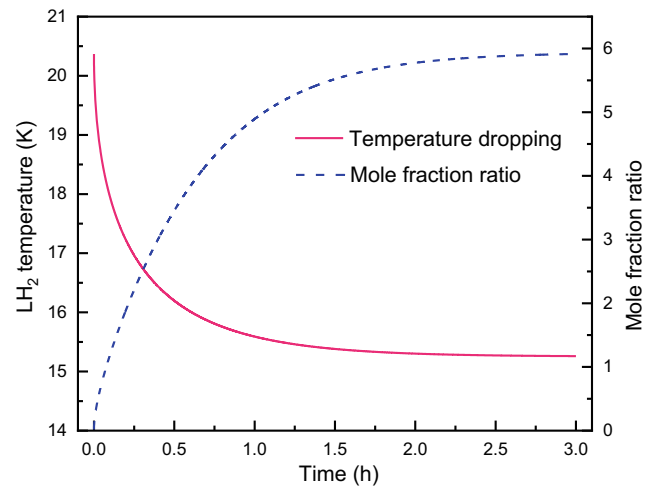
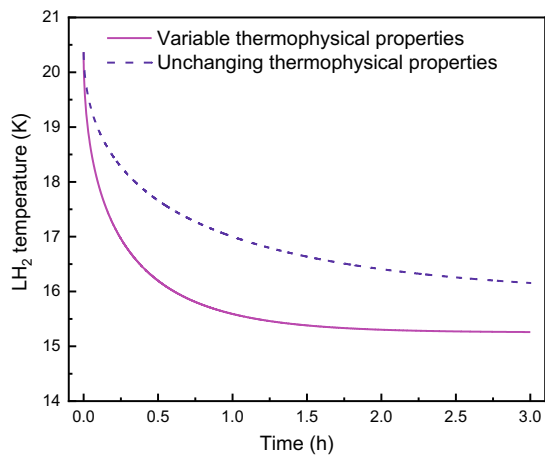
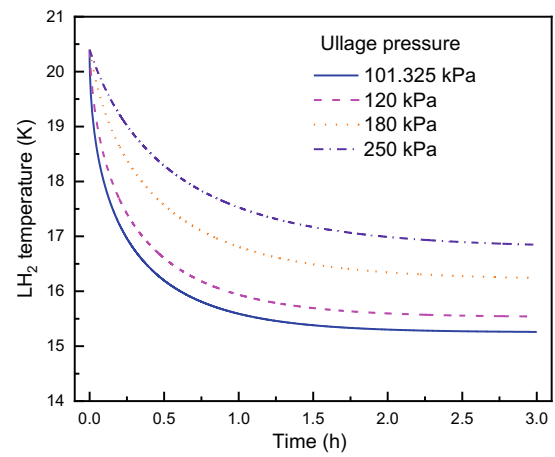


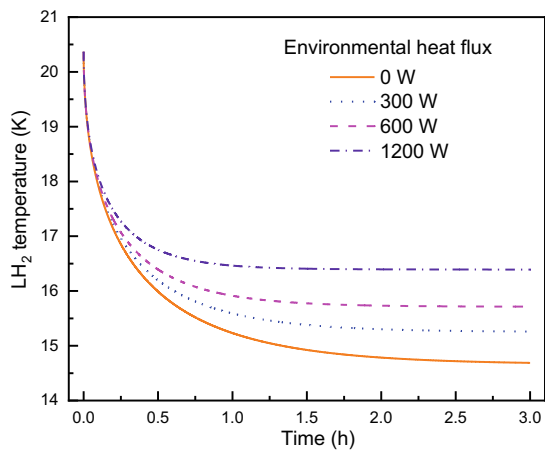
Fig. 12 Profiles of temperature and mole fraction ratio



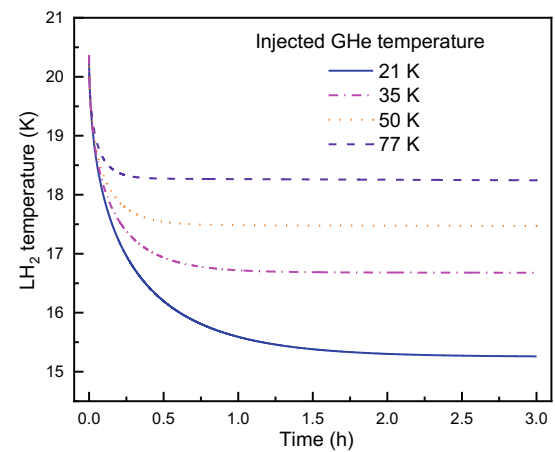
(a) effect of thermophysical properties in the cooling process



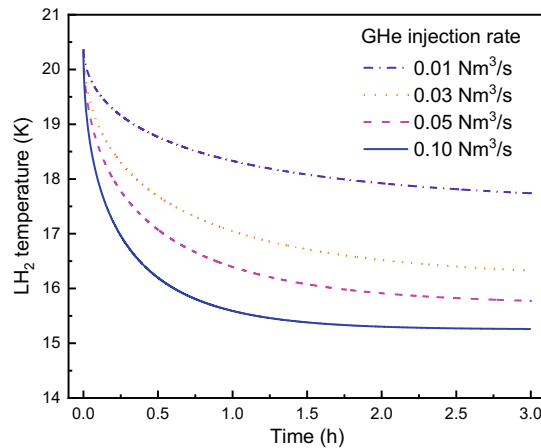
(b) effect of ullage pressure in the cooling process



(c) effect of environmental heat flux in the cooling process



(d) effect of GHe injection temperature in the cooling process



(e) effect of GHe injection rate in the cooling process

Fig. 13 Influence law of different factors on LH₂ temperature cooling

For the hydrostatic pressure, the criterion in the design of the engineering is that the time to reach phase equilibrium for a

single bubble should be equal to the time of floating, thus the appropriate liquid level is calculated.

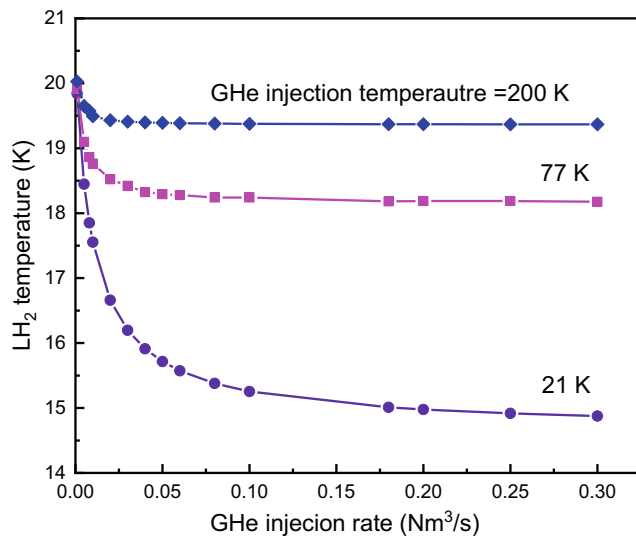


Fig. 14 Effects of GHe injection rate and temperature on minimum attainable temperature

4.3 LH₂ cooling process

Table 5 provides the input conditions on thermodynamic model analysis. The results are presented in Figs. 12, 13, 14, 15, 16, 17 and 18.

Figure 12 shows the change of temperature and mole fraction ratio. From this figure it is observed that the LH₂ temperature goes down rapidly with the continuous injection GHe, and gradually becomes flat at 15.5 K. After 3 h, the LH₂ temperature eventually drops to 15.2 K. Because the corresponding saturation pressure decreases with the decline of LH₂ temperature, the capacity of mass transfer weakens, resulting in the decrease of liquid evaporation rate. In addition, the profile of mole fraction ratio also illustrates this point. As seen in this figure, the mole fraction ratio gradually increases as the LH₂ temperature declines. In other words, the quantity

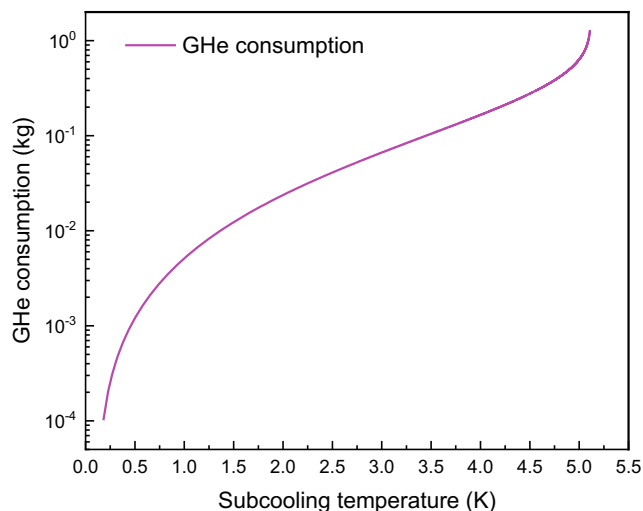


Fig. 15 Relation between GHe consumption and subcooling temperature for unit mass LH₂

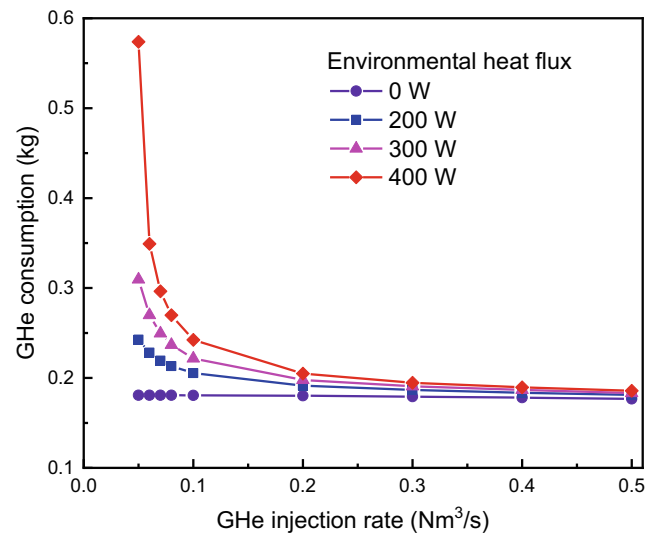


Fig. 16 Effects of environmental heat flux and GHe injection rate on GHe consumption

of taking away hydrogen vapor through 1 mol helium gas becomes less and less with the decreasing LH₂ temperature.

Figure 13 displays the influence of different factors on LH₂ cooling. As shown in Fig. 13 (a), it is obviously observed that the thermophysical properties have significant effect on the calculated results. The maximum relative error considering change of thermophysical properties reaches by 9.3%. Figure 13b reflects the effect of ullage pressure. It can be seen that when the injection time is a constant, the subcooling degree goes down with the increase of ullage pressure, while the subcooling degree is a constant, the time needed is shorter. When the ullage pressure increases by 2.5 times, the minimum attainable temperature increases by 1.6 K. It means that the cooling effect becomes poor. Since the partial pressure of GHe goes up with the increasing ullage pressure, the mass transfer of hydrogen molecules weakens.

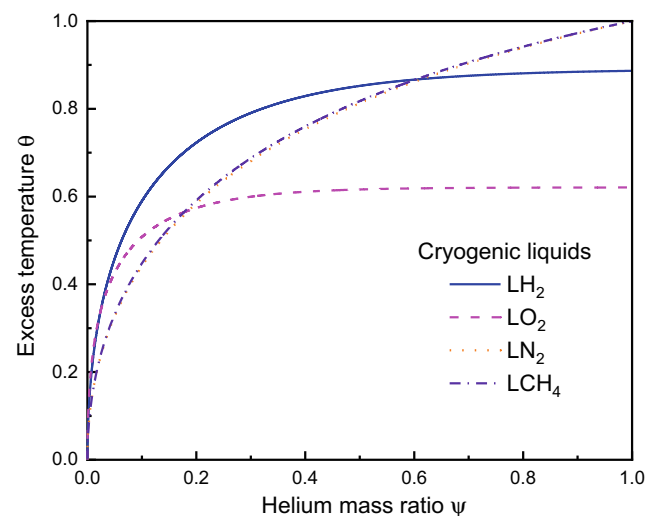


Fig. 17 Effect of cryogenic liquids on the minimum attainable temperature

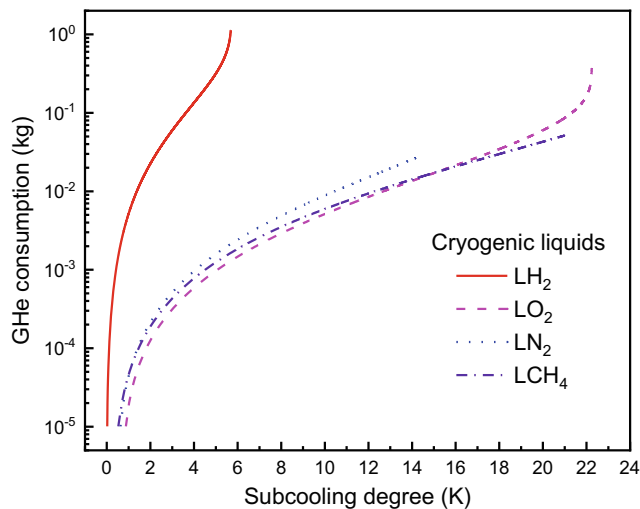


Fig. 18 Effect of cryogenic liquids with varying subcooling degree on the GHe consumption

Figure 13c and d present the effects of environmental heat flux and injected GHe temperature on the cooling effect. From these two figures, it is clearly seen that the environmental heat flux and the injected GHe temperature have slight influence on temperature dropping of LH_2 before 0.1 h, while the cooling effect gradually becomes remarkable in the cooling latter period. Finally, the higher the environmental heat flux and injected GHe temperature, the smaller the temperature dropping of LH_2 . That is to say, the minimum attainable temperature goes up by 1.7 K, when the environmental heat flux increases from 0 W to 1200 W. When the injected GHe temperature increases from 21 to 77 K, the minimum attainable temperature increases by 2.98 K. It means that the cooling effect of LH_2 becomes very poor, and GHe consumption increases, since more extra heat is brought into the cooling system.

Figure 13e indicates that the GHe injection rate has remarkable influence on LH_2 cooling. The higher the GHe injection rate, the greater the temperature dropping of LH_2 . When the GHe injection rate increases by 10 times, the subcooling degree increases by 2.95 K. Because more hydrogen vapor is taken away, the cooling effect of LH_2 becomes better with the increase of the GHe injection rate.

4.4 Minimum attainable temperature

Figure 14 illustrates the effects of GHe injection rate and temperature on minimum attainable temperature. At the GHe injection rate in excess of approximately $0.18 \text{ Nm}^3/\text{s}$, the subcooling temperature would reach 15 K when the GHe injection temperature is precooled with LH_2 , and the temperature would reach 18.2 K with liquid nitrogen precooling. From this figure, it can be also clearly seen that the cooling down rate can be accelerated considerably by increasing the GHe injection rate.

4.5 GHe consumption

Figure 15 displays the relation between GHe consumption and subcooling degree. From this figure, it is clearly observed that when unit mass LH_2 is cooled down, the GHe consumption firstly goes up exponentially as the subcooling degree increases, and then begins to level off. That is to say, when the LH_2 temperature drops to a certain extent, the capacity of obtaining subcooling degree by continuing to inject GHe into the unit mass LH_2 has already been limited badly. Taking the cost into account, there will be an optimal subcooling degree, i.e., the lowest subcooling degree could be obtained with the least cost.

Figure 16 presents the effects of environmental heat flux and GHe injection rate on GHe consumption at unit mass LH_2 obtaining 4.3 K subcooling degree. From this figure, it is clear that the GHe consumption sharply falls with the increase of GHe injection rate and the decline of environment heat flux, and then the falling rate slowly becomes flat. Consequently, the low GHe injection rate is not recommended to cool LH_2 at higher environmental heat flux.

4.6 Cooling effect of other cryogenic liquids

To further study the cooling capacity of various cryogenic liquids using the injected helium gas method, two dimensionless parameters are defined as

$$\theta = \frac{T_{\text{boil}} - T}{T_{\text{boil}} - T_{\text{triple}}} \quad (46)$$

$$\psi = \frac{m_a}{m_n} \quad (47)$$

where θ is the excess temperature; T_{boil} is the normal boiling temperature; T_{triple} is the triple point temperature; ψ is the helium mass ratio; m_a is the required helium consumption at cryogenic liquids cooling to any temperature; m_n is the required helium consumption at cryogenic liquids cooling to the minimum attainable temperature.

The minimum attainable temperature and GHe consumption are discussed, as shown in Figs. 17 and 18. To compare the cooling effect for various cryogenic liquids, many parameters including the mass, the ullage pressure, and the GHe injection rate are set to constants, the environmental heat flux is 0 W, and the injected GHe temperature is respective normal boiling temperature. Figure 17 displays the comparing results of cooling capacity of GHe injected into LH_2 , liquid oxygen (LO_2), liquid methane (LCH_4), and liquid nitrogen (LN_2). From this figure, it is observed that the LCH_4 and LN_2 can reach the triple-point temperature using the method of injected helium gas bubbles, and the maximum attainable excess

temperature of LO_2 is the smallest, i.e., the cooling effect at the near triple-point temperature using that method is not ideal for LO_2 . Since the triple point pressure of LO_2 is the smallest among them, and is very low, its mass transfer capability at the near triple point temperature is extremely weak. In practical engineering, it is not recommended that the cooling method of injection helium bubbles be used to obtain subcooling of LO_2 at the near triple-point temperature.

Figure 18 presents the GHe consumption profiles with varying subcooling degree for various cryogenic liquids of unit mass. It is found that the GHe consumption rapidly goes up with the increase of subcooling degree, and then gradually trends to flat. Given a subcooling degree, the GHe consumptions for various cryogenic liquids of unit mass are different. For instance, when obtaining 4 K subcooling degree, the GHe consumption for unit mass LH_2 is the largest, while it is the least for unit mass LO_2 .

5 Conclusions

The cooling characteristics of GHe injected into LH_2 are investigated by using the simultaneous heat and mass transfer model and the thermodynamic model proposed in the present paper, and many influence factors are discussed. The following conclusions may be drawn:

- (1) The cooling method of using GHe bubbling to obtain larger subcooling degree of LH_2 is feasible and effective. It is found that the cooling rate can be accelerated by increasing the GHe injection rate, quickly reducing the LH_2 temperature to a desired level. The cooling effect becomes better with the decreases of ullage pressure, injected GHe temperature and environmental heat flux.
- (2) During the helium injection process, the heat transfer can be treated as an instantaneous process, while the mass transfer process needs time to meet the requirement of phase equilibrium. In the design of a cooling device, the mass transfer process for GHe injected into LH_2 should be taken into account as far as possible to reach the concentration equilibrium, thereby reducing the helium consumption.
- (3) The calculated results using the thermodynamic model are in good agreement with the experimental data. It is found that both GHe injection rate and environmental heat flux have significant influences on the GHe consumption, i.e., the GHe consumption falls with the increase of GHe injection rate and the decrease of environment heat flux. Consequently, the suitable GHe injection rate is recommended to cool LH_2 at higher environmental heat flux.
- (4) It is also feasible that the cooling method of GHe bubbling can be used for other cryogenic liquids except liquid hydrogen, while the cooling effects are different. It is found that the triple-point temperature of LN_2 and LCH_4 can easily be reached, but the LH_2 and LO_2 can be hardly cooled down to their freezing point, especially for the LO_2 at the minimum attainable temperature being far away from its triple-point temperature. The GHe consumption is the largest for LH_2 cooling at the requirement of a certain subcooling degree among the four cryogenic liquids.

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